

Experiment 'p_4_3_d_count_vowels' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_3_d_count_vowels

Natural language statement: null

Method signature: p_4_3_d_count_vowels(s: string) returns (count: int)

Ensures

- `count == count_vowels(s)`

Functional Code Given

```
function count_vowels(s: string): int {  
  if s == "" then 0  
  else if s[0] in ['a','e','i','o','u','A','E','I','O','U'] then count_vowels(s[1..]) + 1  
  else count_vowels(s[1..])  
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_3_d_count_vowels(s: string)` returns `(count: int)`

The method should respect the following contract:

`ensures count == count_vowels(s)`

The contract uses the following dafny code:

```
function count_vowels(s: string): int {
  if s == "" then 0
  else if s[0] in ['a','e','i','o','u','A','E','I','O','U'] then count_vowels(s[1..]) + 1
  else count_vowels(s[1..])
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnLsPEaCc8puF001rzt9bD03sRAaN

Model: gpt-5.5-2026-04-23

Created at: 1780654537

Finish reason: stop

Usage: 444 tokens in, and 1991 tokens out

```
{
  if s == "" {
    count := 0;
  } else {
    var rest := p_4_3_d_count_vowels(s[1..]);
    if s[0] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] {
      count := rest + 1;
    } else {
      count := rest;
    }
  }
}
```

```
}
```

Final Program

```
function count_vowels(s: string): int {
  if s == "" then 0
  else if s[0] in ['a','e','i','o','u','A','E','I','O','U'] then count_vowels(s[1..]) + 1
  else count_vowels(s[1..])
}

method p_4_3_d_count_vowels(s: string) returns (count: int)
  ensures count == count_vowels(s)
{
  if s == "" {
    count := 0;
  } else {
    var rest := p_4_3_d_count_vowels(s[1..]);
    if s[0] in ['a','e','i','o','u','A','E','I','O','U'] {
      count := rest + 1;
    } else {
      count := rest;
    }
  }
}
```

Total Token Usage

Input tokens: 444

Output tokens: 1991

Reasoning tokens: 1847

Sum of ‘total tokens’: 2435

Experiment Timings

Overall Experiment started at 1780654536650, ended at 1780654605840, lasting 69190ms (69.19 seconds)

Iteration #1 started at 1780654536669, ended at 1780654605840, lasting 69171ms (69.17 seconds)